

NEEDS ANALYSIS

Tailgate Talk 2.0: Preparing Road Crews for Technology-Assisted Drivers

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Project Overview

This is a proof-of-concept microlearning module developed to demonstrate instructional design process against a real, current strategic priority. It is a portfolio piece, not an actual work product: the strategic and regulatory context below is drawn from public materials and general road-construction safety regulation, but the specific field procedures depicted (traffic control plan confirmation steps, driver-assist and navigation-app awareness questions) are illustrative and modeled on general industry practice, not any municipality's actual internal procedures, which I don't have access to as a candidate.

Business / Strategic Driver

Some cities' safety action plans name the elimination of serious injuries and fatalities on roads as a standing council priority, with annual reporting tracking progress toward that goal. At the same time, the vehicle fleet on the roads is changing: a growing share of vehicles come equipped with adaptive cruise control, lane-keeping assist, and automatic emergency braking, and drivers increasingly route themselves using real-time navigation apps rather than direct observation alone. Both trends are good news for road safety in general. They also introduce a hazard a work-zone tailgate talk needs to catch.

This is a well-documented, industry-wide pattern not unique to any one city: as vehicle automation and navigation technology spread, road crews face a genuinely new hazard category. An approaching vehicle's own systems, not just its driver's attention, increasingly help determine whether it slows for a temporary closure, and a freshly set-up closure may not yet appear in a driver's navigation app at all. Existing safety-communication habits, built around the assumption that a human driver who sees the crew will slow down for them, need an explicit update to keep pace with the road environment itself.

Performance Problem / Gap Statement

The gap is not that road crews don't know how to run a tailgate talk or set up a compliant work zone because, under the Occupational Health and Safety Act and Ontario Traffic Manual Book 7, that process (traffic control plan confirmation, hazard review, protective measures, roles) is already thoroughly trained and regulated. The gap is that the standard tailgate-talk checklist hasn't explicitly caught up to a road where an increasing share of approaching vehicles are only partly under direct driver control: without a deliberate update, an experienced crew's well-practiced habits can carry a "if the driver can see us, the driver will stop for us" assumption that no longer holds for every vehicle on the road.

Target Audience

Audience	Prior Knowledge	Gap This Module Addresses
Primary: field crew / road maintenance workers and Traffic Control Persons (TCPs)	Fully trained on OHSA and Ontario Traffic Manual Book 7 work-zone setup and	Has not yet had the tailgate-talk checklist explicitly updated for driver-assist

	tailgate-talk procedures; multiple years of field experience	technology and navigation-app blind spots as vehicle automation spreads
Secondary: field supervisors who lead briefings	Same baseline as above, plus responsibility for running the briefing and confirming crew understanding	Needs to model and enforce the updated question set, not just personally know it

Task / Content Analysis

Standard tailgate talk today	Updated: Tailgate Talk 2.0
1. Confirm traffic control plan and closure details with supervisor	1. Confirm traffic control plan and closure details with supervisor (unchanged)
2. Review hazards specific to the job	2. Review hazards specific to the job (unchanged)
3. Confirm protective measures (high-visibility apparel, cone/sign placement)	3. Confirm protective measures (high-visibility apparel, cone/sign placement) (unchanged)
4. Confirm each person's role	4. Confirm each person's role (unchanged)
— (not currently a standard step)	5. NEW: Confirm: has this closure been reported so it appears in real time for GPS/navigation apps, and is this a corridor where many vehicles run adaptive cruise control or lane-keep assist? If yes, add advance warning signage and/or a second Traffic Control Person rather than relying on standard buffer distance alone.

Learning Objectives

Written to the ABCD (Audience–Behavior–Condition–Degree) format used throughout an Ontario city's posting language around measurable, results-driven outcomes:

1. Given a job-site scenario involving a work zone on a corridor with driver-assist-equipped traffic or an unreported closure, the learner will correctly identify that an additional confirmation question is required before work begins, with no incorrect identifications on the graded scenario.
2. The learner will state, in their own words, the two new hazard categories introduced by vehicle automation and navigation technology (driver-assist systems not reliably recognizing temporary traffic control; closures not yet reflected in navigation apps) with 100% accuracy on the knowledge check.
3. The learner will apply the updated tailgate-talk job aid to a simulated pre-job briefing, correctly sequencing hazards, protective measures, roles, and the new driver-assist check.
4. (Transfer objective, evaluated on the job, not in the module) The learner will use the Tailgate Talk 2.0 job aid on their next work order involving a work zone on a higher-speed or high-traffic corridor.

Delivery Recommendation

On-demand, mobile-enabled microlearning built in Articulate Rise 360, matching the tools and delivery modes named directly in the posting. Estimated learner runtime: 10–12 minutes, short enough to assign ahead of a seasonal ramp-up in road work without pulling crews off the schedule. The experiential core is a branching scenario simulating a real tailgate-talk conversation, so learners practice the judgment call, not just recall a fact. A companion one-page PDF job aid is designed for printing and keeping in the truck, since field connectivity for a mobile device can't be assumed.

Constraints

- Built as a proof-of-concept scoped to one short module rather than a full training program.
- No access to any city's actual internal traffic control procedures, SME review, or brand assets; content is deliberately built on public strategic materials and general industry/regulatory practice, and is framed as illustrative throughout.
- Single-person production (no dedicated videographer or voice talent).

Evaluation Plan (Kirkpatrick-Aligned)

Level	In this proof-of-concept	In a real deployment
1. Reaction	N/A (single-audience demo)	Post-module micro-survey: relevance and clarity ratings
2. Learning	Knowledge check + scenario completion, built into the module	Same, tracked via LMS/SCORM at scale
3. Behavior	Job aid designed for field use; not field-tested here	Spot audits of tailgate talks on higher-risk corridors
4. Results	N/A	Trend in near-miss/incident reports tied to work-zone intrusions by driver-assist-equipped vehicles

Sources Consulted

- City of Ottawa, Road Safety Action Plan 2020–2024, Final Report (Council documents, ottawa.ca): <https://pub-ottawa.escribemeetings.com/filestream.ashx?documentid=27362>
- City of Ottawa, Strategic Road Safety Action Plan, Annual Report (Council documents): <https://pub-ottawa.escribemeetings.com/filestream.ashx?DocumentId=215979>
- Ontario Traffic Manual, Book 7 — Temporary Conditions (Ministry of Transportation of Ontario): <https://otc.org/wp-content/uploads/2023/01/OTM-Book-7-Temporary-Conditions-Office-Edition-April-2022.pdf>
- General framework: Occupational Health and Safety Act (Ontario); general transportation-safety literature on connected/automated vehicles and real-time work-zone data feeds for navigation systems: <https://www.ontario.ca/laws/statute/90o01>